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HRT – STUDY 01 · WORKING PAPER 1.0
26 OCTOBER 2026

The 2026–27 El Niño and crop risk, read from open data

Relative Niño-3.4, satellite rainfall and four decades of national yields for fifteen crop belts

HORTUS RESEARCH, LASTFOMO · DATA TO 24 OCTOBER 2026

BACKGROUND By October 2026 the Pacific had warmed past every month in the satellite record on the relative Niño-3.4 index. Agency outlooks describe the likely impacts in words; growers, buyers and aid planners also need crop-by-crop numbers that anyone can check and update.

AIM To measure, from open data only, where the 2026–27 El Niño puts harvests at risk, by how much, and with what confidence.

METHODS We rebuilt the relative Niño-3.4 index from NOAA OISST v2.1 (January 1982 to 24 October 2026). For fifteen crop belts we regressed seasonal rainfall (NASA IMERG, 2001–2025) and maximum temperature (MERRA-2, 1982–2025) on the index averaged over the same months, and tested MERRA-2 rainfall against IMERG. For 23 crop–country pairs we regressed detrended FAO yields (1982–2018) on the crop-season index, repeated each fit under six detrending choices and without the four very strong events, and derived 80% prediction intervals for the 2026 and 2027 harvests.

RESULTS The index reached +2.67 °C in October 2026 (preliminary), above the previous monthly high of +2.51 °C (November 2015). Seasonal rain fell with the index in 12 of 15 belts, significantly in 7 (Java −97% per °C; eastern Australia −43%; Central America's Dry Corridor −16%), and rose significantly in the Argentine Pampas (+19%), southern Brazil (+16%) and Kenya's short rains (+25%). MERRA-2 and IMERG barely agreed over the Zambezi belt and Ethiopia (r = 0.08 or less); MERRA-2 put the 2023–24 Zambezi season near normal where IMERG measured −53%. South African maize yields fell 13.3% per °C (95% CI −19.4 to −7.2) under every detrending choice. World yields showed no response. At a season-mean index of 2.5 °C the models give South African maize a 92% chance of a loss larger than 10% in 2027 (mean −38%; 80% interval −63% to −13%).

CONCLUSIONS The 2026–27 risk is regional: Southern Africa's 2027 maize harvest first, then the Dry Corridor and the dry side of the Indo-Pacific. Scenarios above about 2.1 °C extrapolate beyond the record, and over Africa the choice of rainfall dataset can reverse a conclusion.

KEYWORDS ENSO · relative Niño index · teleconnections · maize · IMERG · yield forecasting

1. Introduction

El Niño shifts the tropical rain belts for a year or more, and farms on the dry side of the shift lose harvests. Global studies show that El Niño and La Niña move the yields of maize, rice, wheat and soybean across

large parts of the world's cropland, with losses in some regions and gains in others (Iizumi et al., 2014). Droughts and extreme heat cut national cereal production by 9–10% between 1964 and 2007, through smaller harvested areas as well as lower yields (Lesk et al., 2016). The economic damage lasts: Callahan and Mankin (2023) attribute global income losses of US$4.1 trillion and US$5.7 trillion to the 1982–83 and 1997–98 events.

How strong an event is depends on how it is measured. The classic Niño-3.4 anomaly mixes El Niño with the warming of the whole tropical ocean; a relative index subtracts the tropical mean and rescales the result (van Oldenborgh et al., 2021; L'Heureux et al., 2024). NOAA's Climate Prediction Center adopted this relative index operationally from 1 February 2026 (National Weather Service, 2026). On 8 October 2026 CPC kept its El Niño Advisory in force and favoured a strong to very strong event through January–March 2027 (Climate Prediction Center, 2026). FEWS NET (2026) named Southern Africa and unimodal East Africa as the regions of highest concern for 2027 food assistance needs.

Those outlooks are probabilistic about the ocean and qualitative about crops. What they do not give is a reproducible, crop-by-crop estimate that a grower, a buyer or a planner can rerun each month as data arrive. This study builds one from open data and asks four questions:

1. **RQ1.** How strong is the 2026–27 event on the relative index, and how does it compare with past events?
2. **RQ2.** Which crop belts lose or gain seasonal rain in El Niño years, and has 2026 followed the pattern so far?
3. **RQ3.** How sensitive are national yields to the index, and how robust is that sensitivity?
4. **RQ4.** What do the historical relationships imply for the 2026 and 2027 harvests, and how uncertain is that?

2. Background: from ocean to field

The links run through the atmosphere. When warm water spreads east along the equatorial Pacific, the rising air of the Walker circulation follows it; rain shifts with the air, leaving some regions dry and others wet. Ropelewski and Halpert (1987) mapped the regions where this signal is consistent, using records from more than 1,700 stations. The link is not fixed: Kumar et al. (2006) showed that Indian monsoon failures depend on where the Pacific warming is centred, with central-Pacific warming more effective at producing drought than warming in the far east.

Rain matters to a crop only at certain stages. A dry spell at maize flowering costs more than the same deficit at emergence, so a seasonal index is a summary, not a crop model. National yields add a further filter: they average dry and wet districts, irrigated and rain-fed fields. A yield regression on an ENSO index therefore measures the historical net effect on a national harvest; it does not explain the mechanism. We use it because it is the effect that matters to supply, and because every input is open and current to within weeks.

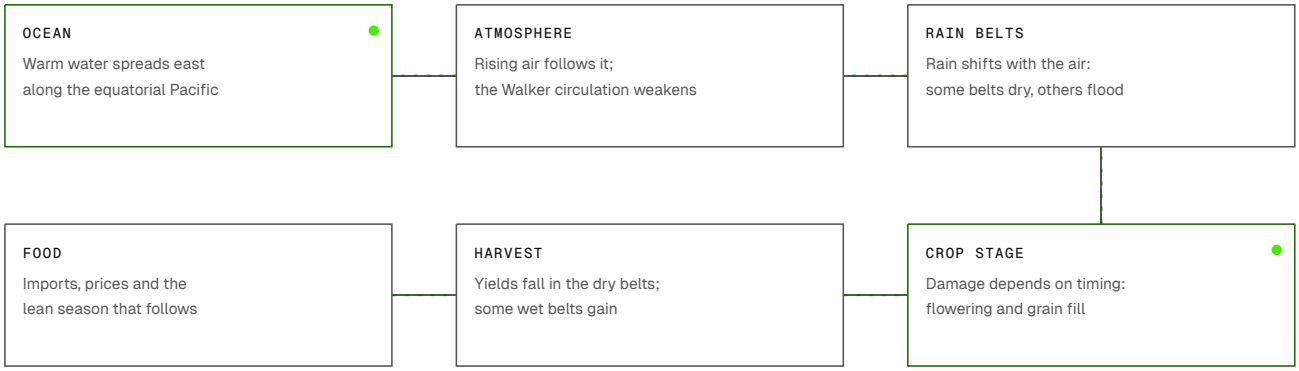


FIGURE 1. How El Niño reaches a field. After Ropelewski and Halpert (1987) and Iizumi et al. (2014).

3. Data and methods

3.1 RELATIVE NIÑO-3.4

We read NOAA OISST v2.1 daily sea-surface temperature (0.25°; Huang et al., 2021) from the NOAA archive on the AWS Open Data registry, three days per month (the 5th, 15th and 25th, or the nearest available day), January 1982 to 24 October 2026. October 2026 rests on preliminary files. For each day we computed cosine-latitude-weighted means over Niño-3.4 (5° S–5° N, 170°–120° W) and over the tropical ocean (20° S–20° N, all longitudes), took monthly means and anomalies from the 1991–2020 monthly climatology, and formed the relative index as the Niño-3.4 anomaly minus the tropical anomaly, rescaled so that its 1991–2020 standard deviation equals that of the Niño-3.4 anomaly (scale factor 1.172). This follows the method of van Oldenborgh et al. (2021) and L'Heureux et al. (2024); it is a reconstruction, not CPC's official series. An El Niño event is five or more consecutive overlapping three-month means at or above +0.5 °C; we class events by their three-month peak as weak (0.5–1.0 °C), moderate (1.0–1.5), strong (1.5–2.0) and very strong (2.0 or more).

3.2 CROP BELTS, RAINFALL AND HEAT

We defined fifteen boxes over the main crop belts on the El Niño front line, each with the months that matter to its main crop (Table 1). Regional means of monthly rainfall from NASA IMERG and of rainfall, mean daily maximum temperature and root-zone wetness from the MERRA-2 reanalysis (Gelaro et al., 2017) came from the NASA POWER archive on AWS, weighted by the cosine of latitude, with MERRA-2 restricted to land cells. Seasonal rain is expressed as a percentage of the 2001–2020 mean (IMERG) or the 1991–2020 mean (MERRA-2); heat as the anomaly of the mean daily maximum from 1991–2020. For each belt we regressed seasonal rain on the relative index averaged over the same months: IMERG for 2001–2025 (25 seasons), MERRA-2 for 1982–2025 (44 seasons). We report the slope, its 95% confidence interval and p-value, and the correlation between the two datasets' seasonal anomalies over 2001–2025. Seasons that had begun by October 2026 were scored with the complete months to September against the same months' normal.

3.3 YIELDS

National yields for 1961–2018 are FAO data as processed by Our World in Data; World Bank cereal yields extend the check to 2024. We detrended each series with robust LOWESS (Cleveland, 1979; span 0.3, three robustness iterations) and expressed each year as a percentage above or below trend. Each crop–country pair was matched to a belt and season (Table 2); where the season runs into the new year (Southern Hemisphere summer crops, Kenya's short rains, West African cocoa), the harvest is counted in the following year. We regressed the yield anomaly on the crop-season index for seasons beginning 1982–

2018 (n = 36 or 37) and tested robustness under six detrending choices (LOWESS spans 0.2, 0.3 and 0.5; linear; quadratic; five-year centred moving average) and with the very strong seasons (1982, 1991, 1997, 2015) left out. World yields were regressed on the November–January index.

3.4 OUTLOOK

For seasons that had begun by October 2026, the predictor is the index observed so far in the season. For seasons still ahead, we used scenarios of 2.0, 2.5 and 3.0 °C for the season mean; the live outlook in the Hortus research pages takes any value from 1.5 to 3.0 °C. From each regression we derived the mean prediction, the 80% prediction interval for a single new season, and the probability of a loss larger than 10%, from the t distribution of the prediction error with n – 2 degrees of freedom. We flag every prediction beyond the strongest season in its own record.

3.5 EVIDENCE LABELS

Each main finding carries a label: STRONG EVIDENCE where results are consistent across datasets, methods or many seasons; MODERATE where one solid dataset or several analyses agree; LIMITED where results rest on few seasons, one method or extrapolation.

4. Results

4.1 THE EVENT

The relative index reached +2.67 °C in October 2026 (1–24 October, three sampled days, preliminary) and +2.33 °C in September. No month since 1982 came close: the previous highest was +2.51 °C in November 2015. The unadjusted Niño-3.4 anomaly was larger still, +3.40 °C against 1991–2020, because the whole tropical ocean is warmer than its normal. By October, 2026 runs ahead of 1982 and 1997 at the same stage (both +2.21 °C) and well ahead of 2015 (+1.85 °C). The three-month mean had already reached +2.37 °C in the season centred on September, level with the peak of 1982–83 and above the peaks of 1991–92, 1997–98 and 2015–16 (Figure 2). STRONG EVIDENCE

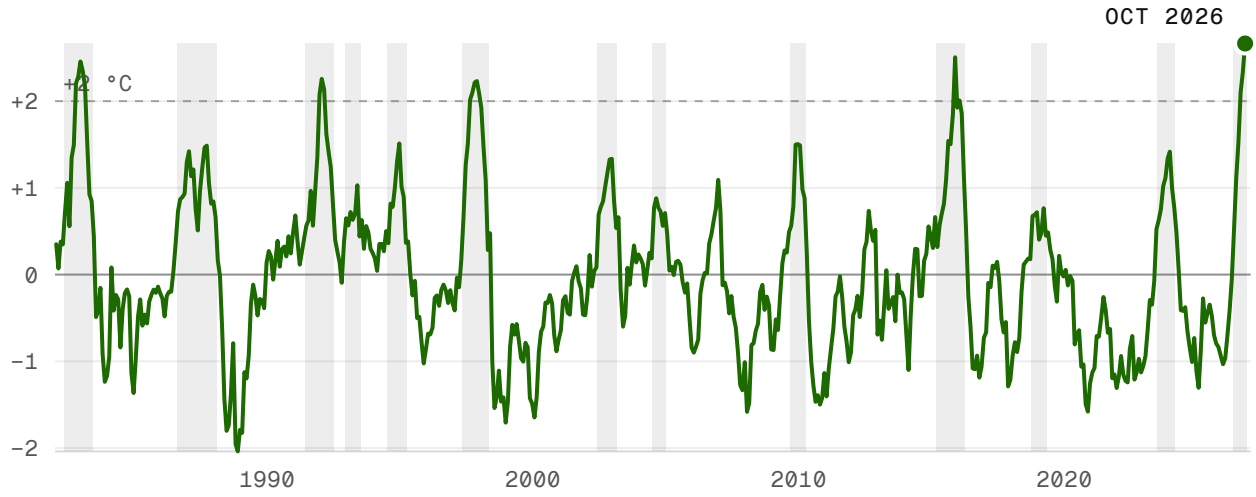


FIGURE 2. Relative Niño-3.4, monthly, 1982–2026, reconstructed from NOAA OISST v2.1. Shaded: El Niño events. The last value covers 1–24 October 2026 and is preliminary.

4.2 RAIN MOVES

Seasonal rain fell with the index in 12 of the 15 belts, significantly at 5% in seven (Table 1; Figure 3). The response is largest where the normal season is dry: Java's July–October rain falls by 97% of its normal for each degree of the index, eastern Australia's June–October rain by 43%. Central America's Dry Corridor

(–16%), northern and eastern Ethiopia (–13%), India's core monsoon zone (–9%), the South African Highveld (–9%) and Thailand's rice plains (–8%) also dry out. Three belts get wetter: Kenya's short rains (+25%), the Argentine Pampas (+19%) and southern Brazil (+16%). The same seasons run hotter on the dry side: the mean daily maximum over both Southern African belts rises by 0.87 °C per °C of the index ($p < 0.001$), and over the Dry Corridor by 0.59 °C ($p = 0.007$), while the Pampas runs 0.63 °C cooler. STRONG EVIDENCE

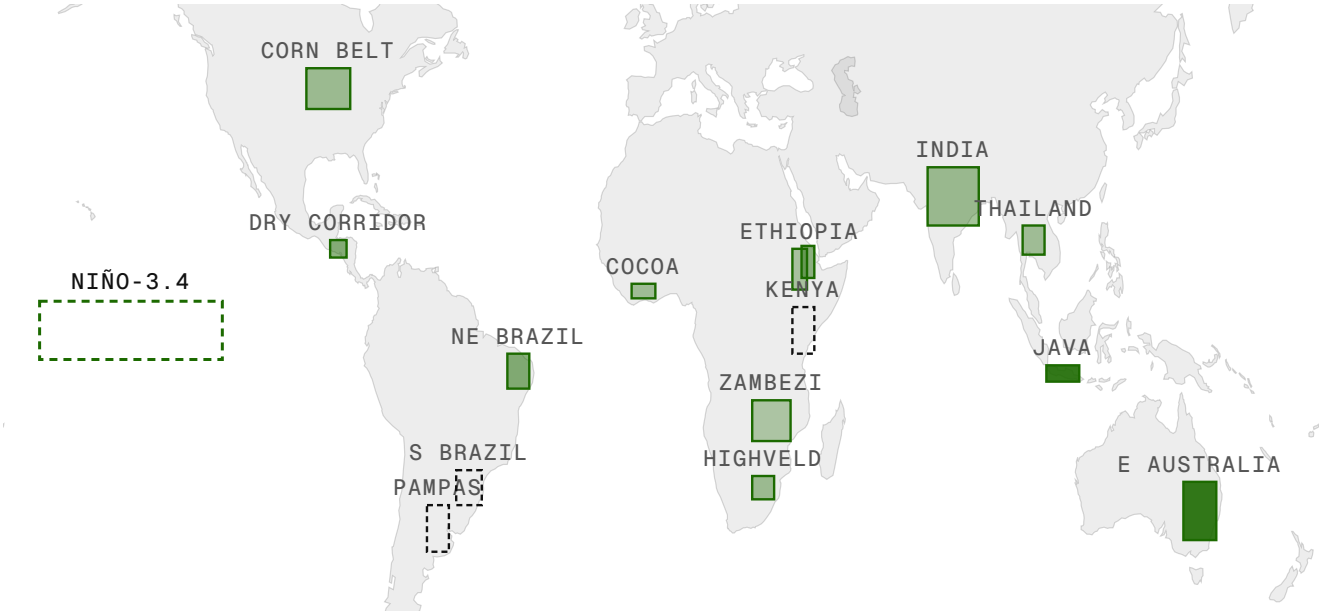


FIGURE 3. Crop belts and their rainfall response to the relative index (IMERG, 2001–2025). Filled boxes dry out in El Niño seasons; dashed boxes get wetter; shading scales with the response.

TABLE 1. Crop belts, seasons and rainfall response. IMERG 2001–2025 (n = 25); MERRA-2 1982–2025 (n = 44); heat from MERRA-2.

BELT	CROP	SEASON	RAIN, % PER °C (95% CI)	P	MERRA-2	HEAT, °C PER °C
South Africa Highveld	Maize	Dec–Mar	−8.6 (−16 to −1)	0.030	−9.0	+0.87
Zambezi maize belt (ZMB, ZWE, MWI)	Maize	Dec–Mar	−3.5 (−13 to +6)	0.447	−12.2	+0.87
Ethiopian highlands (kiremt)	Maize, teff, wheat	Jun–Sep	−5.1 (−11 to +1)	0.096	−12.3	+0.33
Northern and eastern Ethiopia	Sorghum, maize, teff	Jun–Sep	−13.3 (−21 to −5)	0.003	−16.7	+0.43
India core monsoon zone	Rice (kharif)	Jun–Sep	−9.4 (−15 to −4)	0.003	−6.2	+0.32
Java	Rice	Jul–Oct	−96.7 (−138 to −55)	<0.001	−64.7	+0.09
Thailand rice plains	Rice	May–Oct	−7.5 (−13 to −2)	0.010	−9.1	+0.28
Eastern Australia wheat belt	Wheat	Jun–Oct	−42.6 (−65 to −20)	<0.001	−26.0	+0.35
Central America Dry Corridor	Maize, beans	May–Oct	−15.6 (−24 to −7)	<0.001	−22.8	+0.59
Côte d'Ivoire–Ghana cocoa belt	Cocoa	Nov–Mar	−7.6 (−18 to +3)	0.138	−9.7	+0.36
North-east Brazil	Maize, beans	Feb–May	−16.5 (−43 to +10)	0.208	−26.1	+0.80
Argentine Pampas	Soybean, maize	Nov–Feb	+19.1 (+12 to +26)	<0.001	+13.0	−0.62
Southern Brazil	Soybean	Nov–Feb	+15.8 (+8 to +23)	<0.001	+13.6	−0.31
Kenya and southern Somalia short rains	Maize	Oct–Dec	+25.1 (+5 to +45)	0.014	+20.3	−0.39
US Corn Belt	Maize, soybean	Jun–Aug	−8.4 (−21 to +4)	0.188	+0.5	−0.56

4.3 A BLIND SPOT IN THE DATA

We first ran the rainfall analysis with MERRA-2 alone, because it reaches back to 1980. Over the Zambezi maize belt it put the 2023–24 season near normal (–0.9%), while IMERG measured –52.8%; that season led Zambia, Zimbabwe and Malawi to declare national drought disasters (World Food Programme, 2024). Across 2001–2025 the two datasets' seasonal anomalies barely correlate over the Zambezi belt (r = 0.02), the Ethiopian highlands (r = –0.13) and northern and eastern Ethiopia (r = 0.08); elsewhere they agree closely (r ≥ 0.89 in nine belts; Figure 4). MERRA-2 corrects its precipitation with gauge-based products whose coverage is thin over much of Africa (Reichle et al., 2017). We therefore used IMERG for all rainfall results and MERRA-2 for temperature only. STRONG EVIDENCE

The swap has a cost. IMERG starts in 2000, so its 25 seasons contain one very strong event (2015–16). Over the Zambezi belt the IMERG slope is weak and uncertain (–3.6% per °C, 95% CI –13% to +6%), while MERRA-2's 44 seasons give –12.2% ($p < 0.001$) but cannot be trusted there. The yield record, which integrates the rain that actually fell, is the better witness for that belt (§4.5).

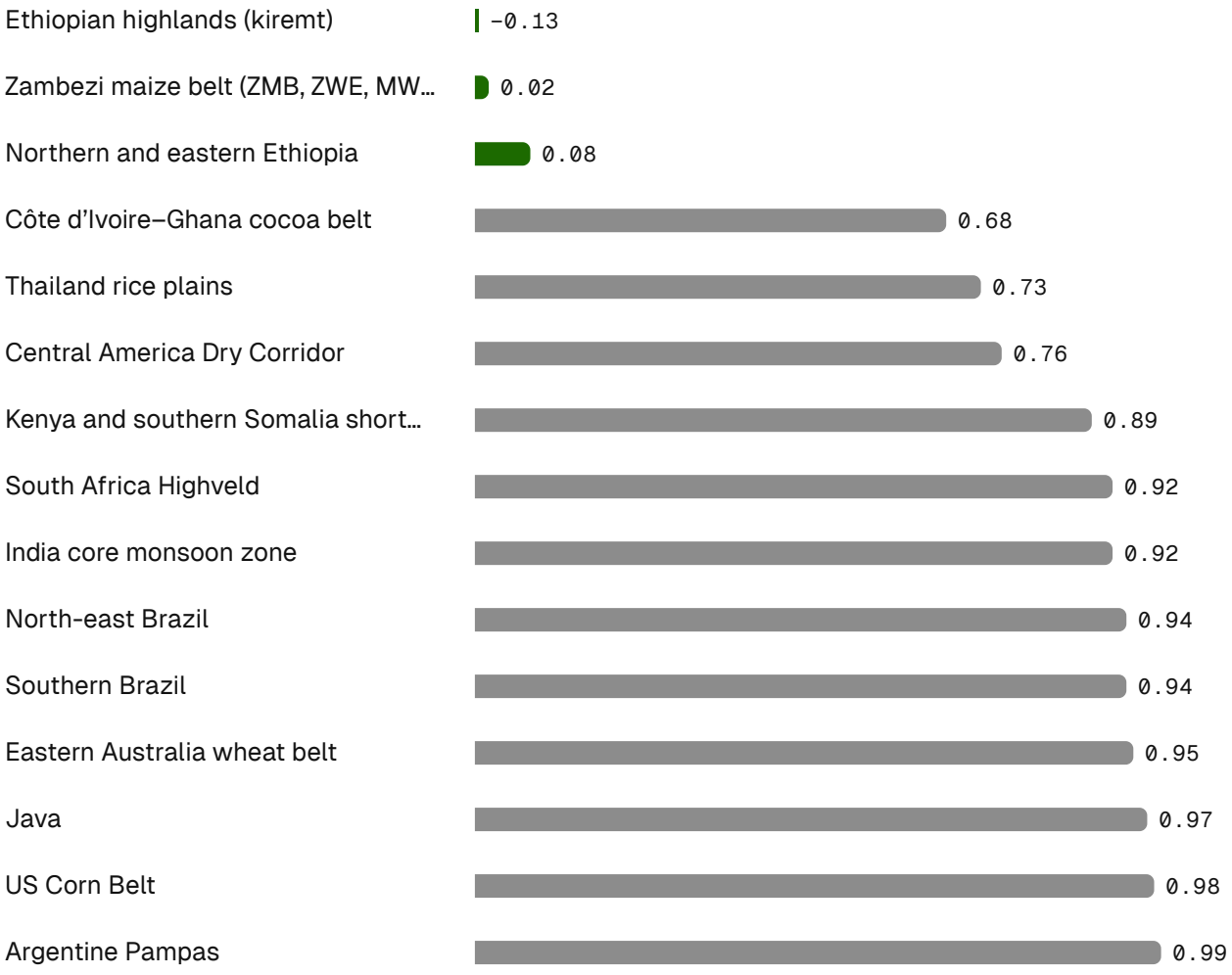


FIGURE 4. Correlation between MERRA-2 and IMERG seasonal rainfall anomalies, 2001–2025. Highlighted: belts where the datasets disagree enough to change conclusions (r below 0.5).

4.4 2026 SO FAR

Every belt whose season had begun shows the El Niño pattern in rain or heat (Figure 5). Java's July–September rain was 85% below normal and the Dry Corridor's May–September rain 27% below. Thailand's rice plains were 20% below, northern and eastern Ethiopia 19% below and India's core monsoon zone 12% below, close to the India Meteorological Department's national figure of 759.4 mm against a normal of

868.6 mm (–12.6%; DSIJ, 2026). The dry belts were also hot: northern and eastern Ethiopia +2.7 °C, the Ethiopian highlands +2.5 °C and the Dry Corridor +2.4 °C above the normal mean daily maximum. The highlands, with near-normal rain (+2%), suffered heat rather than drought. MODERATE EVIDENCE (one season, partly preliminary data)

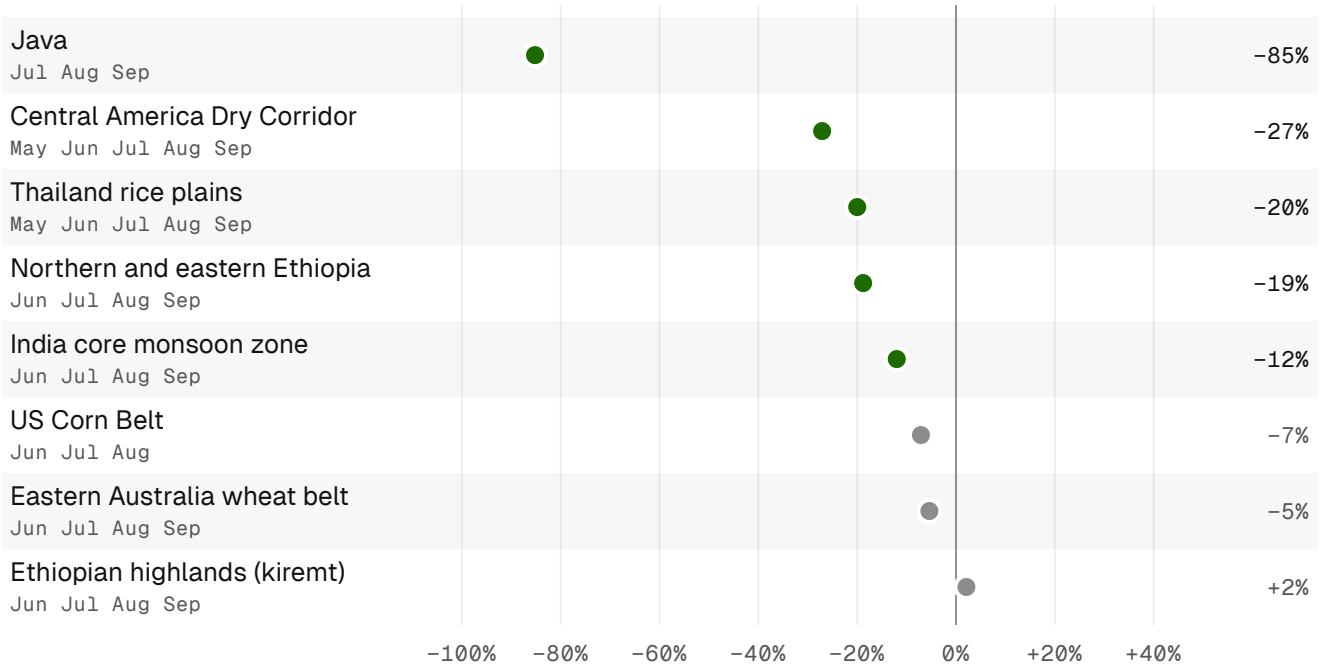


FIGURE 5. Rain in the 2026 season to the end of September, % of the 2001–2020 normal for the same months (IMERG).

4.5 YIELDS

Southern Africa's maize carries the largest and best-measured signal (Table 2; Figure 6). South African yields fall 13.3% below trend for each degree of the crop-season index (95% CI –19.4 to –7.2; $p < 0.001$), Zimbabwe's 12.2% ($p = 0.02$) and Malawi's 8.9% ($p = 0.02$); Mozambique's –9.1% narrowly misses significance ($p = 0.06$) and Zambia's –3.5% does not approach it. In the harvests after the four very strong events, 16 of the 20 Southern African maize yields fell below trend, and the 1992 harvest fell 54–72% below trend in all five countries. Elsewhere, El Salvador's maize (–6.0%) and India's rice (–2.3%) respond significantly; Argentina's maize gains 4.7% per °C ($p = 0.003$) and United States soybeans 3.2% ($p = 0.048$). STRONG EVIDENCE for South Africa; MODERATE for the others

The South African result survives every robustness test: the slope stays between –12.8% and –14.1% under all six detrending choices ($p \leq 0.002$), and remains negative without the very strong seasons (–8.0% to –11.4%; significant in four of six fits). Malawi is significant in five of six fits, Zimbabwe in four; both depend on the very strong seasons, without which neither is significant. India's rice keeps its slope (–1.5% to –2.5%) with or without them. The World Bank cereal series, which runs to 2024, confirms the pattern: South Africa –10.2% per °C (95% CI –14.4 to –5.9), with the 2024 harvest 17.7% below trend after a moderate event, and Mozambique becomes significant (–9.8%, $p = 0.01$).

World yields barely move. None of the global maize, wheat, rice or soybean series responds significantly in the onset year or the next (every slope between –1.0% and +0.9% per °C; smallest $p = 0.13$), in line with FEWS NET's observation that global output has changed little on balance in strong and very strong El Niño years since 1980 (FEWS NET, 2026). The global average hides the households inside the dry belts. STRONG EVIDENCE

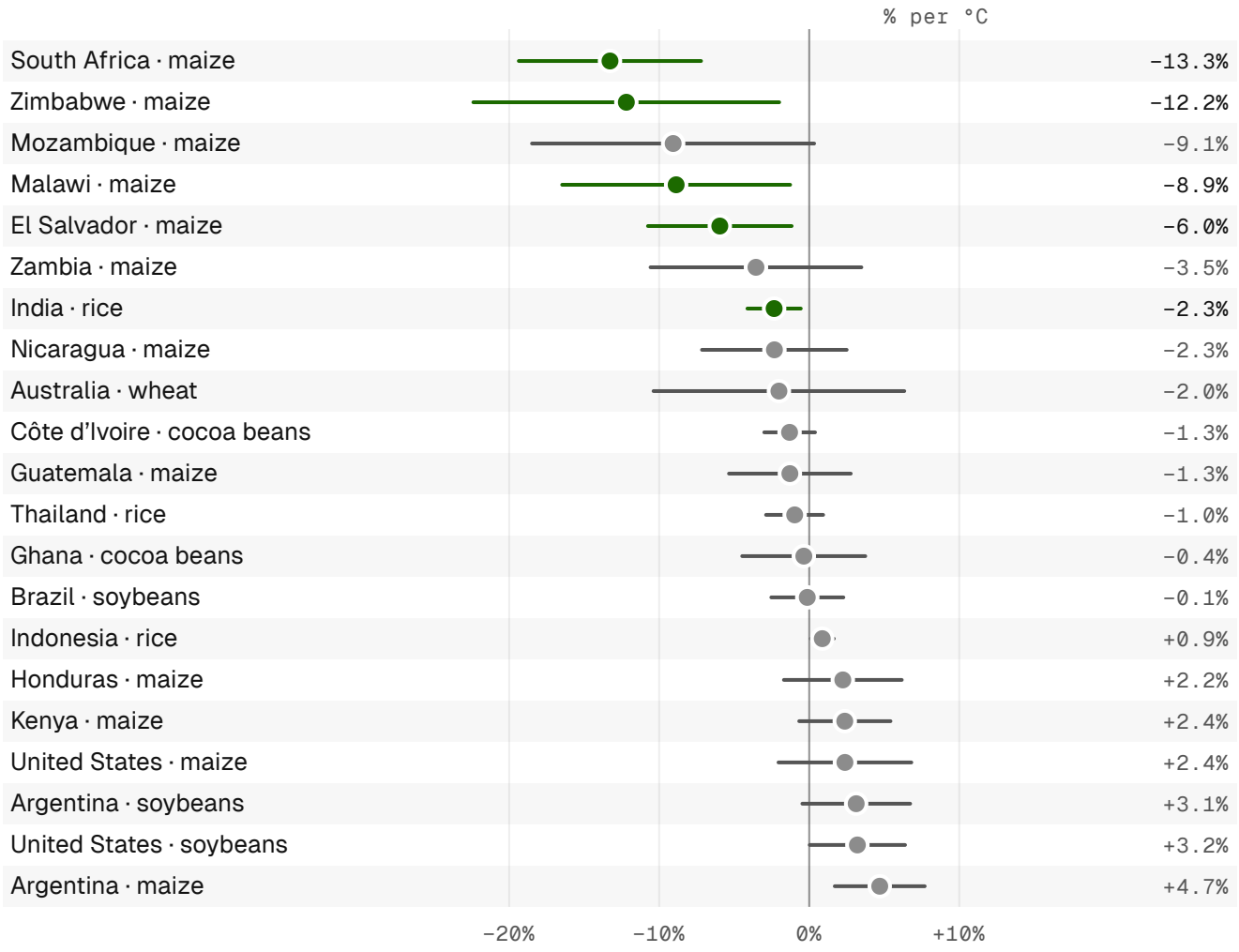


FIGURE 6. Yield response to the crop-season relative index, % of trend per °C, with 95% confidence intervals (FAO, 1982–2018). Highlighted: significant losses.

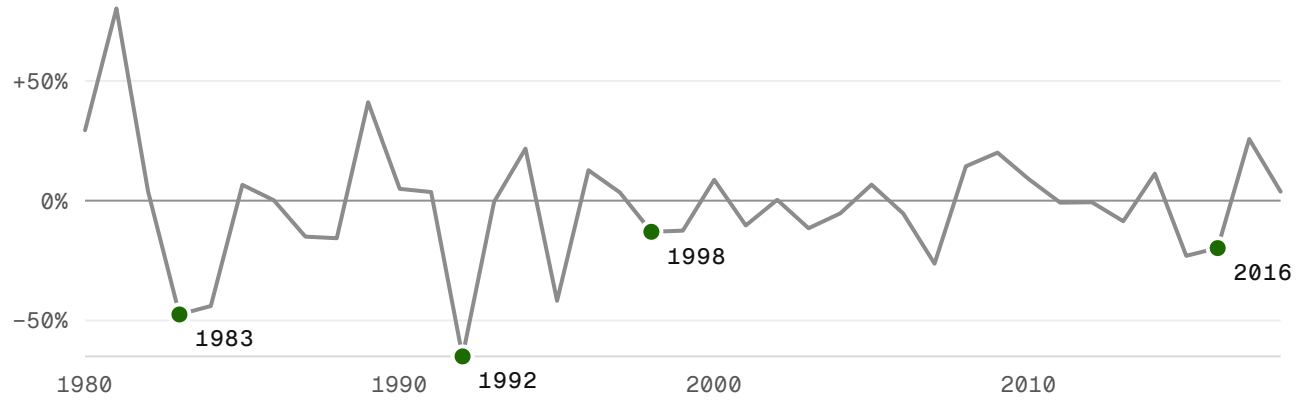


FIGURE 7. South African maize yield, % above or below trend, 1980–2018. Marked: harvests after the four very strong events.

TABLE 2. Yield sensitivity and robustness. Slope: % of trend per °C (95% CI). Fits: detrending choices (of six) significant at 5%. Without VS: slope range with the very strong seasons removed.

COUNTRY · CROP	HARVEST	N	SLOPE (95% CI)	P	FITS	WITHOUT VS
South Africa · maize	next year	36	−13.3 (−19.4 to −7.2)	<0.001	6/6	−11.4 to −8.0
Zimbabwe · maize	next year	36	−12.2 (−22.4 to −2.0)	0.021	4/6	−8.3 to −2.5

COUNTRY · CROP	HARVEST	N	SLOPE (95% CI)	P	FITS	WITHOUT VS
Mozambique · maize	next year	36	−9.1 (−18.5 to +0.3)	0.058	0/6	−12.0 to −2.0
Malawi · maize	next year	36	−8.9 (−16.5 to −1.3)	0.024	5/6	−8.2 to −2.7
El Salvador · maize	same year	37	−6.0 (−10.8 to −1.2)	0.016	6/6	−5.4 to −1.8
Zambia · maize	next year	36	−3.5 (−10.6 to +3.5)	0.312	0/6	+1.6 to +4.1
India · rice	same year	37	−2.3 (−4.1 to −0.6)	0.011	5/6	−2.5 to −1.8
Nicaragua · maize	same year	37	−2.3 (−7.2 to +2.5)	0.336	—	—
Australia · wheat	same year	37	−2.0 (−10.4 to +6.3)	0.626	0/6	+0.3 to +1.0
Côte d'Ivoire · cocoa beans	next year	36	−1.3 (−3.0 to +0.4)	0.127	—	—
Guatemala · maize	same year	37	−1.3 (−5.3 to +2.8)	0.523	—	—
Thailand · rice	same year	37	−1.0 (−2.9 to +0.9)	0.306	—	—
Ghana · cocoa beans	next year	36	−0.4 (−4.5 to +3.7)	0.858	—	—
Brazil · soybeans	next year	36	−0.1 (−2.5 to +2.3)	0.916	—	—
Indonesia · rice	same year	37	+0.9 (+0.1 to +1.6)	0.029	—	—
Honduras · maize	same year	37	+2.2 (−1.7 to +6.2)	0.255	—	—
Kenya · maize	next year	36	+2.4 (−0.7 to +5.4)	0.122	—	—
United States · maize	same year	37	+2.4 (−2.0 to +6.8)	0.283	—	—
Argentina · soybeans	next year	36	+3.1 (−0.5 to +6.7)	0.086	—	—
United States · soybeans	same year	37	+3.2 (+0.0 to +6.4)	0.048	—	—
Argentina · maize	next year	36	+4.7 (+1.7 to +7.7)	0.003	5/6	+4.9 to +6.3

4.6 OUTLOOK FOR 2026 AND 2027

At a season-mean index of 2.5 °C, the regressions put the 2027 maize harvest 38% below trend in South Africa (80% prediction interval −63% to −13%; probability of a loss above 10%: 92%) and 36% below in Zimbabwe (−77% to +6%; 78%), with Malawi (−22%; 69%) and Mozambique (−22%; 66%) next (Table 3; Figure 8). For seasons already under way, El Salvador's 2026 maize is the most exposed (−11%; 55%), followed by Australia's wheat (−9%; 47%). Argentina's 2027 maize is the clearest gainer (+12%; 2% chance of a loss above 10%). LIMITED EVIDENCE: every Southern African scenario at 2.5 °C lies beyond the strongest season in the 1982–2018 record (2.12 °C), and for eight of the ten seasons already under way the index observed so far also exceeds the record, so most of these means extrapolate a straight line past the data.

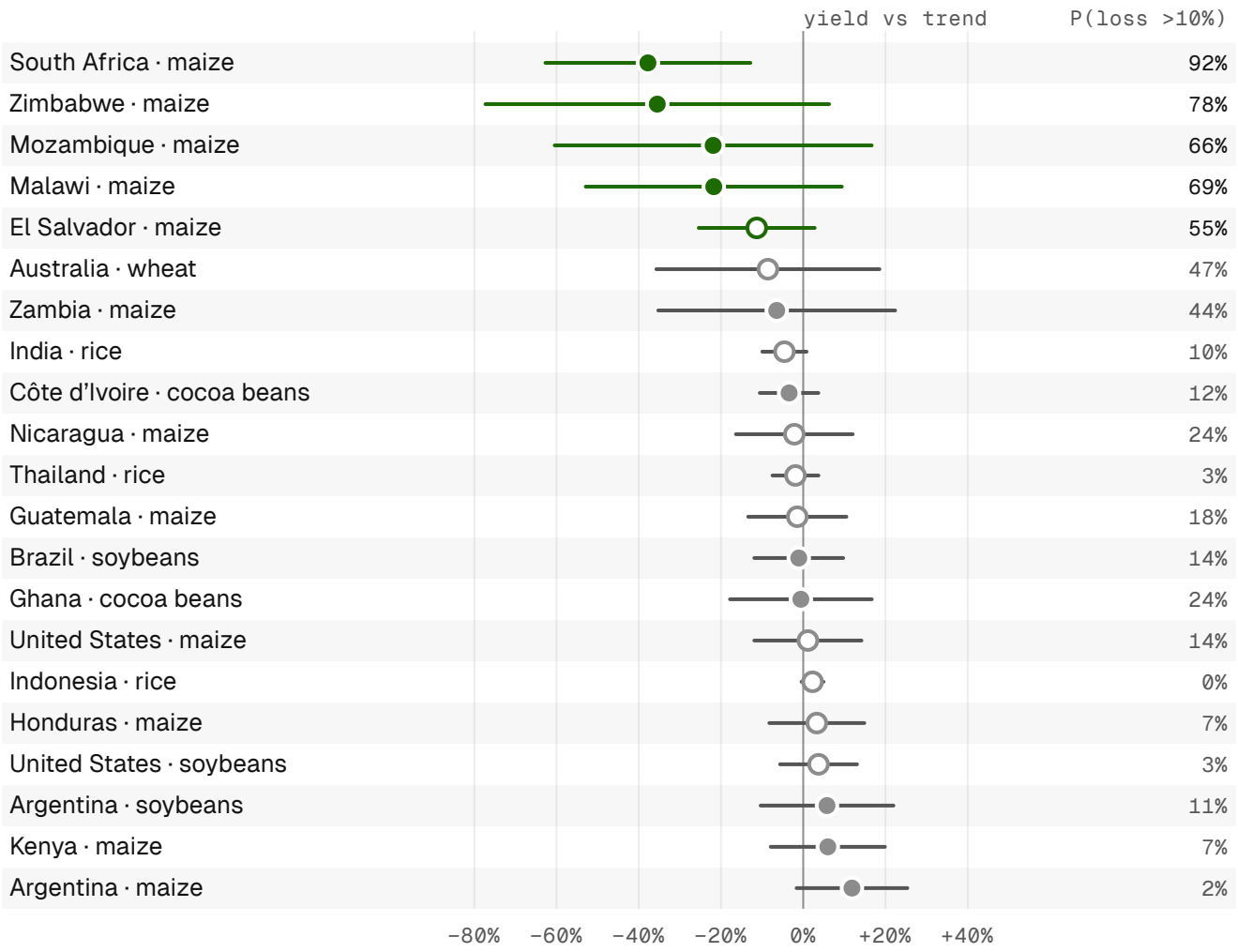


FIGURE 8. Expected yield against trend at a season-mean index of 2.5 °C (seasons ahead) or the index observed so far (seasons under way), with 80% prediction intervals.

TABLE 3. Outlook by crop and country. Index: the scenario (2.5 °C) or the mean observed so far. P: probability of a yield loss larger than 10%.

CROP · COUNTRY	HARVEST	INDEX, °C	MEAN, %	80% INTERVAL, %	P
Maize · South Africa	2027	2.50*	−38	−63 to −13	92%
Maize · Zimbabwe	2027	2.50*	−36	−77 to +6	78%
Maize · Malawi	2027	2.50*	−22	−53 to +9	69%
Maize · Mozambique	2027	2.50*	−22	−60 to +17	66%
Maize · El Salvador	2026	1.71*	−11	−25 to +3	55%
Wheat · Australia	2026	1.95*	−9	−36 to +19	47%
Maize · Zambia	2027	2.50*	−6	−35 to +22	44%
Maize · Nicaragua	2026	1.71*	−2	−16 to +12	24%
Cocoa beans · Ghana	2027	2.50*	−1	−18 to +17	24%
Maize · Guatemala	2026	1.71*	−1	−13 to +11	18%
Soybeans · Brazil	2027	2.50*	−1	−12 to +10	14%
Maize · United States	2026	1.58	+1	−12 to +14	14%
Cocoa beans · Côte d'Ivoire	2027	2.50*	−3	−11 to +4	12%
Soybeans · Argentina	2027	2.50*	+6	−10 to +22	11%

CROP · COUNTRY	HARVEST	INDEX, °C	MEAN, %	80% INTERVAL, %	P
Rice · India	2026	1.77*	−5	−10 to +1	10%
Maize · Honduras	2026	1.71*	+3	−8 to +15	7%
Maize · Kenya	2027	2.50*	+6	−8 to +20	7%
Rice · Thailand	2026	1.71*	−2	−7 to +4	3%
Soybeans · United States	2026	1.58	+4	−6 to +13	3%
Maize · Argentina	2027	2.50*	+12	−2 to +25	2%
Rice · Indonesia	2026	2.16*	+2	−0 to +5	0%

* Beyond the strongest crop-season index in the 1982–2018 record for that pair.

5. Discussion

RQ1. On the relative index the 2026–27 event is already the strongest monthly reading since 1982, and CPC expects it to remain strong to very strong through early 2027. The unadjusted anomaly (+3.40 °C) overstates it by mixing in tropical-wide warming; the relative index is the fairer comparison with 1982, 1997 and 2015, and it is the index CPC now uses.

RQ2. The rainfall pattern is the textbook one: the Maritime Continent, eastern Australia, southern Africa, Central America and the Indian monsoon dry out, south-eastern South America and equatorial East Africa get wetter (Ropelewski & Halpert, 1987). What the satellite record adds is size. Java's dry-season rain nearly vanishes in strong events, and in 2026 it did: −85% to September. Our own Sentinel-2 images of West Java show the result on the ground, with the share of rice land reaching green canopy in the dry season falling from 91% in 2025 to 62% in 2026 (Study 02).

RQ3. National yields carry the signal where the crop is rain-fed and the season coincides with the dry phase: Southern African maize, Central American maize and Indian kharif rice. Where crops are irrigated or the season falls outside the teleconnection, as with Indonesian rice or US maize, the national yield barely moves, even in a year like 2026. Indonesia's small positive slope does not mean drought helps: droughts reduce harvested area as well as yields (Lesk et al., 2016), and a national yield cannot see a field that was never planted. Production and imports, not yield alone, decide what reaches a table.

RQ4. The outlook ranks risk more reliably than it sizes it. Southern African maize stands out under any reasonable index value; the exact loss does not. A straight line fitted to 1982–2018 cannot know whether losses saturate or accelerate beyond 2.1 °C, and the very strong events are only four. Read the means as direction and rough size, the intervals as the honest width of history, and the probabilities as a ranking.

LIMITATIONS

- **Beyond the record.** Scenarios above 2.1–2.3 °C extrapolate past every season in the yield sample.
- **Nations, not fields.** National yields average many districts; the boxes summarise where a crop grows, not where it is planted this year.
- **Yield is not supply.** Area planted, stocks, trade and prices decide availability; this study does not model them.
- **One ocean.** The Indian Ocean Dipole and Atlantic temperatures can amplify or offset El Niño; we do not include them.
- **Preliminary data.** October 2026 sea temperatures, recent IMERG months and the latest yields will be revised.

- **Many tests.** We report 38 main regressions without correcting for multiple comparisons; at 5%, about two could be significant by chance. The findings we label strong hold across datasets or methods.
- **Our index is a reconstruction.** It samples three days a month and uses OISST rather than CPC's inputs, so monthly values will differ somewhat from the official series.

6. Recommendations

- **Agencies and donors** should plan now for Southern Africa's 2027 lean season, pre-positioning maize and cash support. STRONG
- **Growers in the dry belts** facing December–March planting should consider shorter-season maize and save water early; growers in the Pampas and southern Brazil should plan for wet harvests. MODERATE
- **Buyers** should expect Southern Africa to import maize in 2027 if these losses materialise, and price for a regional, not a global, shortfall. MODERATE
- **Analysts** should check reanalysis rainfall against satellite estimates before using it over Africa. STRONG

7. Conclusion

From open data alone, and within days of the month's end, it is possible to say where an El Niño will cost harvests and roughly how much. In 2026–27 the answer is Southern Africa's 2027 maize first, then the Dry Corridor, Java and the other dry belts of the Indo-Pacific, while world totals barely move. The same pipeline can be rerun each month as the season unfolds; the Hortus research pages hold the live version.

Data availability. All inputs are public: NOAA OISST v2.1 daily files (AWS Open Data, noaa-cdr-sea-surface-temp-optimum-interpolation-pds); NASA POWER monthly and daily archives of MERRA-2 and IMERG (AWS, nasa-power); FAO crop yields as published by Our World in Data (key crop yields, 1961–2018); World Bank indicator AG.YLD.CREL.KG; Natural Earth coastlines. Data were read on 26 October 2026.

Code availability. The processing scripts are published with the Hortus research pages (code/study-01).

Use of AI tools. Data retrieval, analysis code, figures and the first draft of this text were produced with Claude (Anthropic), an AI model, working for the Hortus team. Every number comes from the published scripts; every reference was checked against an online record on 26 October 2026. The Hortus team is responsible for the content. AI tools are not authors.

Funding and conflicts of interest. No external funding. Hortus sells growing hardware described in Study 02; this study recommends no product.

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